

# Circular Economy Ontology Network (CEON) - Actor Module

## Metadata

|                            |                                                                                                         |
|----------------------------|---------------------------------------------------------------------------------------------------------|
| IRI                        | <a href="http://w3id.org/CEON/ontology/actor/">http://w3id.org/CEON/ontology/actor/</a>                 |
| Title                      | Circular Economy Ontology Network (CEON) - Actor Module                                                 |
| Creator                    | Huanyu Li                                                                                               |
| Contributor                | Eva Blomqvist<br>Mikael Lindecrantz<br>Robin Keskisärkkä                                                |
| Date Created               | 2024-11-13                                                                                              |
| License                    | <a href="https://creativecommons.org/licenses/by/4.0/">https://creativecommons.org/licenses/by/4.0/</a> |
| Version Iri                | <a href="http://w3id.org/CEON/ontology/actor/0.3/">http://w3id.org/CEON/ontology/actor/0.3/</a>         |
| Version Info               | 0.3                                                                                                     |
| Preferred Namespace Prefix | actor                                                                                                   |
| Preferred Namespace Uri    | <a href="http://w3id.org/CEON/ontology/actor/">http://w3id.org/CEON/ontology/actor/</a>                 |
| Description                | The Actor module of CEON (Circular Economy Ontology Network).                                           |

### Covers Requirements

In addition to requirements covered by imported ODPs, covers the following requirements from Onto-DESiDE D3.1: C0-1, C0-2, C3-4, T3-4

In addition to requirements covered by imported ODPs, covers the following requirements from Onto-DESiDE D3.2: C0-1, C0-2, C3-4, T3-4

## Classes

|                             |                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Actor Cvn Role <sup>c</sup> |                                                                                                                                                                                                                                                                                                                                       |
| IRI                         | <a href="http://w3id.org/CEON/ontology/actor/ActorCVNRole">http://w3id.org/CEON/ontology/actor/ActorCVNRole</a>                                                                                                                                                                                                                       |
| Sub Class Of                | <a href="#">ActorCollaborationRole<sup>c</sup></a>                                                                                                                                                                                                                                                                                    |
| Named Individuals           | <a href="#">collector<sup>ni</sup></a><br><a href="#">dismantler<sup>ni</sup></a><br><a href="#">manufacturer<sup>ni</sup></a><br><a href="#">recycler<sup>ni</sup></a><br><a href="#">reseller<sup>ni</sup></a><br><a href="#">seller<sup>ni</sup></a><br><a href="#">supplier<sup>ni</sup></a><br><a href="#">user<sup>ni</sup></a> |

## Actor Collaboration Role<sup>C</sup>

|                              |                                                                                                                                     |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                   | <a href="http://w3id.org/CE0N/ontology/actor/ActorCollaborationRole">http://w3id.org/CE0N/ontology/actor/ActorCollaborationRole</a> |
| <b><u>Sub Class Of</u></b>   | <a href="#">actorODP:Role<sup>C</sup></a>                                                                                           |
| <b><u>Super Class Of</u></b> | <a href="#">ActorCVNRole<sup>C</sup></a><br><a href="#">ActorProcessRole<sup>C</sup></a>                                            |

## Actor Process Role<sup>C</sup>

|                            |                                                                                                                         |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                 | <a href="http://w3id.org/CE0N/ontology/actor/ActorProcessRole">http://w3id.org/CE0N/ontology/actor/ActorProcessRole</a> |
| <b><u>Sub Class Of</u></b> | <a href="#">ActorCollaborationRole<sup>C</sup></a>                                                                      |

## Actor Resource Role<sup>C</sup>

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                      | <a href="http://w3id.org/CE0N/ontology/actor/ActorResourceRole">http://w3id.org/CE0N/ontology/actor/ActorResourceRole</a>                                                                                                                                                                                                                                                                                                                                                               |
| <b><u>Sub Class Of</u></b>      | <a href="#">actorODP:Role<sup>C</sup></a>                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b><u>Named Individuals</u></b> | <a href="#">buyer<sup>ni</sup></a><br><a href="#">consumer<sup>ni</sup></a><br><a href="#">issuer<sup>ni</sup></a><br><a href="#">owner<sup>ni</sup></a><br><a href="#">producer<sup>ni</sup></a><br><a href="#">provider<sup>ni</sup></a><br><a href="#">reseller<sup>ni</sup></a><br><a href="#">seller<sup>ni</sup></a><br><a href="#">supplier<sup>ni</sup></a><br><a href="#">updater<sup>ni</sup></a><br><a href="#">user<sup>ni</sup></a><br><a href="#">viewer<sup>ni</sup></a> |

## Buying Resource<sup>C</sup>

|                               |                                                                                                                                                                |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                    | <a href="http://w3id.org/CE0N/ontology/actor/BuyingResource">http://w3id.org/CE0N/ontology/actor/BuyingResource</a>                                            |
| <b><u>Sub Class Of</u></b>    | <a href="#">actorODP:ResourceRelation<sup>C</sup></a>                                                                                                          |
| <b><u>Equivalentclass</u></b> | <a href="#">actorODP:ResourceRelation<sup>C</sup></a> <i>and</i> <a href="#">actorODP:participantRole<sup>op</sup></a> value <a href="#">buyer<sup>C</sup></a> |

## Cvn Participation<sup>C</sup>

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                 | <a href="http://w3id.org/CE0N/ontology/actor/CVNParticipation">http://w3id.org/CE0N/ontology/actor/CVNParticipation</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b><u>Sub Class Of</u></b> | <a href="#">actorODP:CollaborationParticipation<sup>C</sup></a><br><a href="#">actorODP:participationIn<sup>op</sup></a> some <a href="http://w3id.org/CE0N/ontology/cvn/CVN&lt;sup&gt;C&lt;/sup&gt;">http://w3id.org/CE0N/ontology/cvn/CVN<sup>C</sup></a> <i>and</i><br><a href="#">actorODP:participantRole<sup>op</sup></a> some <a href="#">ActorCVNRole<sup>C</sup></a> <i>and</i><br><a href="#">actorODP:participatingActor<sup>op</sup></a> some <a href="#">actorODP:Actor<sup>C</sup></a> <i>and</i><br><a href="#">actorODP:CollaborationParticipation<sup>C</sup></a> |

## Consuming Resource<sup>C</sup>

|                        |                                                                                                                                                            |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>             | <a href="http://w3id.org/CE0N/ontology/actor/ConsumingResource">http://w3id.org/CE0N/ontology/actor/ConsumingResource</a>                                  |
| <b>Sub Class Of</b>    | <a href="#">actorODP:ResourceRelation<sup>C</sup></a>                                                                                                      |
| <b>Equivalentclass</b> | <a href="#">actorODP:participantRole<sup>OP</sup></a> value <a href="#">consumer<sup>C</sup></a> and <a href="#">actorODP:ResourceRelation<sup>C</sup></a> |

## Issuing Resource<sup>C</sup>

|                        |                                                                                                                                                          |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>             | <a href="http://w3id.org/CE0N/ontology/actor/IssuingResource">http://w3id.org/CE0N/ontology/actor/IssuingResource</a>                                    |
| <b>Sub Class Of</b>    | <a href="#">actorODP:ResourceRelation<sup>C</sup></a>                                                                                                    |
| <b>Equivalentclass</b> | <a href="#">actorODP:participantRole<sup>OP</sup></a> value <a href="#">issuer<sup>C</sup></a> and <a href="#">actorODP:ResourceRelation<sup>C</sup></a> |

## Manufacturing Resource<sup>C</sup>

|                        |                                                                                                                                                                |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>             | <a href="http://w3id.org/CE0N/ontology/actor/ManufacturingResource">http://w3id.org/CE0N/ontology/actor/ManufacturingResource</a>                              |
| <b>Sub Class Of</b>    | <a href="#">actorODP:ResourceRelation<sup>C</sup></a>                                                                                                          |
| <b>Equivalentclass</b> | <a href="#">actorODP:participantRole<sup>OP</sup></a> value <a href="#">manufacturer<sup>C</sup></a> and <a href="#">actorODP:ResourceRelation<sup>C</sup></a> |

## Organisation<sup>C</sup>

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>           | <a href="http://w3id.org/CE0N/ontology/actor/Organisation">http://w3id.org/CE0N/ontology/actor/Organisation</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Is Defined By</b> | ISO 59004:2024 - 3.4.1 organization                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Description</b>   | Person or group of people that has its own functions with responsibilities, authorities, and relationships to achieve its objectives. The concept of organization includes, but is not limited to sole-trader, company, corporation, firm, enterprise, authority, partnership, charity or institution, or part or combination thereof, whether incorporated or not, public or private (e.g. foundation, union, association, agency, municipality, region, country, intergovernmental agencies). A group of organizations can also be considered as an organization that has, alone or collectively, their own objectives. |
| <b>Sub Class Of</b>  | <a href="#">actorODP:Actor<sup>C</sup></a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

## Owning Resource<sup>C</sup>

|                        |                                                                                                                                                         |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>             | <a href="http://w3id.org/CE0N/ontology/actor/OwningResource">http://w3id.org/CE0N/ontology/actor/OwningResource</a>                                     |
| <b>Sub Class Of</b>    | <a href="#">actorODP:ResourceRelation<sup>C</sup></a>                                                                                                   |
| <b>Equivalentclass</b> | <a href="#">actorODP:participantRole<sup>OP</sup></a> value <a href="#">owner<sup>C</sup></a> and <a href="#">actorODP:ResourceRelation<sup>C</sup></a> |

## Person<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/actor/Person>

**Sub Class Of** [actorODP:Actor<sup>C</sup>](#)

## Process Participation<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/actor/ProcessParticipation>

**Sub Class Of** [actorODP:CollaborationParticipation<sup>C</sup>](#)  
[actorODP:CollaborationParticipation<sup>C</sup>](#) *and* [actorODP:participatingActor<sup>OP</sup>](#) some  
[actorODP:Actor<sup>C</sup>](#) *and* [actorODP:participationIn<sup>OP</sup>](#) some  
[http://w3id.org/CE0N/ontology/processODP/Process<sup>C</sup>](http://w3id.org/CE0N/ontology/processODP/Process) *and*  
[actorODP:participantRole<sup>OP</sup>](#) some [ActorProcessRole<sup>C</sup>](#)

## Producing Resource<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/actor/ProducingResource>

**Sub Class Of** [actorODP:ResourceRelation<sup>C</sup>](#)

**Equivalentclass** [actorODP:participantRole<sup>OP</sup>](#) value [producer<sup>C</sup>](#) *and* [actorODP:ResourceRelation<sup>C</sup>](#)

## Providing Resource<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/actor/ProvidingResource>

**Sub Class Of** [actorODP:ResourceRelation<sup>C</sup>](#)

**Equivalentclass** [actorODP:participantRole<sup>OP</sup>](#) value [provider<sup>C</sup>](#) *and* [actorODP:ResourceRelation<sup>C</sup>](#)

## Reselling Resource<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/actor/ResellingResource>

**Sub Class Of** [actorODP:ResourceRelation<sup>C</sup>](#)

**Equivalentclass** [actorODP:ResourceRelation<sup>C</sup>](#) *and* [actorODP:participantRole<sup>OP</sup>](#) value [reseller<sup>C</sup>](#)

## Selling Resource<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/actor/SellingResource>

**Sub Class Of** [actorODP:ResourceRelation<sup>C</sup>](#)

**Equivalentclass** [actorODP:participantRole<sup>OP</sup>](#) value [seller<sup>C</sup>](#) *and* [actorODP:ResourceRelation<sup>C</sup>](#)

## Interested partyStakeholder<sup>C</sup>

|                             |                                                                                                                                                                                                                        |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                  | <a href="http://w3id.org/CE0N/ontology/actor/Stakeholder">http://w3id.org/CE0N/ontology/actor/Stakeholder</a>                                                                                                          |
| <b><u>Is Defined By</u></b> | ISO 59004:2024 - 3.4.2 interested party, stakeholder                                                                                                                                                                   |
| <b><u>Description</u></b>   | Person or organization that can affect, be affected by, or perceive itself to be affected by a decision or activity. To “perceive itself to be affected” means the perception has been made known to the organization. |
| <b><u>Sub Class Of</u></b>  | <a href="#">Organisation</a> <sup>C</sup> or <a href="#">Person</a> <sup>C</sup>                                                                                                                                       |

## Supplying Resource<sup>C</sup>

|                               |                                                                                                                                                               |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                    | <a href="http://w3id.org/CE0N/ontology/actor/SupplyingResource">http://w3id.org/CE0N/ontology/actor/SupplyingResource</a>                                     |
| <b><u>Sub Class Of</u></b>    | <a href="#">actorODP:ResourceRelation</a> <sup>C</sup>                                                                                                        |
| <b><u>Equivalentclass</u></b> | <a href="#">actorODP:ResourceRelation</a> <sup>C</sup> and <a href="#">actorODP:participantRole</a> <sup>OP</sup> value <a href="#">supplier</a> <sup>C</sup> |

## Updating Resource<sup>C</sup>

|                               |                                                                                                                                                              |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                    | <a href="http://w3id.org/CE0N/ontology/actor/UpdatingResource">http://w3id.org/CE0N/ontology/actor/UpdatingResource</a>                                      |
| <b><u>Sub Class Of</u></b>    | <a href="#">actorODP:ResourceRelation</a> <sup>C</sup>                                                                                                       |
| <b><u>Equivalentclass</u></b> | <a href="#">actorODP:participantRole</a> <sup>OP</sup> value <a href="#">updater</a> <sup>C</sup> and <a href="#">actorODP:ResourceRelation</a> <sup>C</sup> |

## Using Resource<sup>C</sup>

|                               |                                                                                                                                                           |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                    | <a href="http://w3id.org/CE0N/ontology/actor/UsingResource">http://w3id.org/CE0N/ontology/actor/UsingResource</a>                                         |
| <b><u>Sub Class Of</u></b>    | <a href="#">actorODP:ResourceRelation</a> <sup>C</sup>                                                                                                    |
| <b><u>Equivalentclass</u></b> | <a href="#">actorODP:participantRole</a> <sup>OP</sup> value <a href="#">user</a> <sup>C</sup> and <a href="#">actorODP:ResourceRelation</a> <sup>C</sup> |

## Viewing Resource<sup>C</sup>

|                               |                                                                                                                                                             |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                    | <a href="http://w3id.org/CE0N/ontology/actor/ViewingResource">http://w3id.org/CE0N/ontology/actor/ViewingResource</a>                                       |
| <b><u>Sub Class Of</u></b>    | <a href="#">actorODP:ResourceRelation</a> <sup>C</sup>                                                                                                      |
| <b><u>Equivalentclass</u></b> | <a href="#">actorODP:ResourceRelation</a> <sup>C</sup> and <a href="#">actorODP:participantRole</a> <sup>OP</sup> value <a href="#">viewer</a> <sup>C</sup> |

## Actor<sup>C</sup>

|                              |                                                                                                         |
|------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                   | <a href="http://w3id.org/CE0N/ontology/actorODP/Actor">http://w3id.org/CE0N/ontology/actorODP/Actor</a> |
| <b><u>Super Class Of</u></b> | <a href="#">Organisation</a> <sup>C</sup><br><a href="#">Person</a> <sup>C</sup>                        |

## Capability<sup>C</sup>

**IRI** <http://w3id.org/CEON/ontology/actorODP/Capability>

**In Domain Of** [capabilityExtent<sup>op</sup>](#)  
[neededResourceRelation<sup>op</sup>](#)

## Collaboration Participation<sup>C</sup>

**IRI** <http://w3id.org/CEON/ontology/actorODP/CollaborationParticipation>

**Super Class Of** [CVNParticipation<sup>C</sup>](#)  
[ProcessParticipation<sup>C</sup>](#)

## Participation<sup>C</sup>

**IRI** <http://w3id.org/CEON/ontology/actorODP/Participation>

## Resource Relation<sup>C</sup>

**IRI** <http://w3id.org/CEON/ontology/actorODP/ResourceRelation>

**Sub Class Of** [actorODP:participantRole<sup>op</sup>](#) some [ActorResourceRole<sup>C</sup>](#) and  
[actorODP:participatingResource<sup>op</sup>](#) some  
[http://w3id.org/CEON/ontology/resourceODP/Resource<sup>C</sup>](http://w3id.org/CEON/ontology/resourceODP/Resource) and  
[actorODP:participatingActor<sup>op</sup>](#) some [actorODP:Actor<sup>C</sup>](#) and  
[actorODP:Participation<sup>C</sup>](#)

**Super Class Of** [BuyingResource<sup>C</sup>](#)  
[ConsumingResource<sup>C</sup>](#)  
[IssuingResource<sup>C</sup>](#)  
[ManufacturingResource<sup>C</sup>](#)  
[OwningResource<sup>C</sup>](#)  
[ProducingResource<sup>C</sup>](#)  
[ProvidingResource<sup>C</sup>](#)  
[ResellingResource<sup>C</sup>](#)  
[SellingResource<sup>C</sup>](#)  
[SupplyingResource<sup>C</sup>](#)  
[UpdatingResource<sup>C</sup>](#)  
[UsingResource<sup>C</sup>](#)  
[ViewingResource<sup>C</sup>](#)

## Role<sup>C</sup>

**IRI** <http://w3id.org/CEON/ontology/actorODP/Role>

**Super Class Of** [ActorCollaborationRole<sup>C</sup>](#)  
[ActorResourceRole<sup>C</sup>](#)

## Cvn<sup>C</sup>

**IRI** <http://w3id.org/CEON/ontology/cvn/CVN>

**Sub Class Of** <http://w3id.org/CEON/ontology/cvn/Collaboration><sup>C</sup>

## Collaboration<sup>C</sup>

**IRI** <http://w3id.org/CEON/ontology/cvn/Collaboration>

**Super Class Of** <http://w3id.org/CEON/ontology/cvn/CVN><sup>C</sup>

## Process<sup>C</sup>

**IRI** <http://w3id.org/CEON/ontology/processODP/Process>

## Resource<sup>C</sup>

**IRI** <http://w3id.org/CEON/ontology/resourceODP/Resource>

## Object Properties

### capability extent<sup>op</sup>

**IRI** <http://w3id.org/CEON/ontology/actor/capabilityExtent>

**Sub Property Of** [actorODP:capabilityProperty](#)<sup>op</sup>

**Domain** [actorODP:Capability](#)<sup>C</sup>

### needed resource relation<sup>op</sup>

**IRI** <http://w3id.org/CEON/ontology/actor/neededResourceRelation>

**Sub Property Of** [actorODP:capabilityProperty](#)<sup>op</sup>

**Domain** [actorODP:Capability](#)<sup>C</sup>

### capability property<sup>op</sup>

**IRI** <http://w3id.org/CEON/ontology/actorODP/capabilityProperty>

**Super Property Of**

- [capabilityExtent](#)<sup>op</sup>
- [neededResourceRelation](#)<sup>op</sup>

### participant role<sup>op</sup>

**IRI** <http://w3id.org/CEON/ontology/actorODP/participantRole>

participating actor<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/actor0DP/participatingActor>

participating resource<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/actor0DP/participatingResource>

participation in<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/actor0DP/participationIn>

## Annotation Properties

contributor<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/contributor>

created<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/created>

creator<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/creator>

description<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/description>

license<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/license>

title<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/title>

preferred namespace prefix<sup>ap</sup>

**IRI** <http://purl.org/vocab/vann/preferredNamespacePrefix>



preferred namespace uri<sup>ap</sup>

**IRI** <http://purl.org/vocab/vann/preferredNamespaceUri>

covers requirements<sup>ap</sup>

**IRI** <http://www.ontologydesignpatterns.org/schemas/cannotation-schema.owl#coversRequirements>

## Namespaces

:

<http://w3id.org/CEON/ontology/actor/>

**actorODP**

<http://w3id.org/CEON/ontology/actorODP/>

**dcterms**

<http://purl.org/dc/terms/>

**odp**

<http://www.ontologydesignpatterns.org/schemas/cannotationschema.owl#>

**owl**

<http://www.w3.org/2002/07/owl#>

**prov**

<http://www.w3.org/ns/prov#>

**rdf**

<http://www.w3.org/1999/02/22-rdf-syntax-ns#>

**rdfs**

<http://www.w3.org/2000/01/rdf-schema#>

**vann**

<http://purl.org/vocab/vann/>

## Legend

|               |                          |
|---------------|--------------------------|
| <sup>c</sup>  | Classes                  |
| <sup>op</sup> | Object<br>Properties     |
| <sup>ap</sup> | Annotation<br>Properties |